

Mergesort

With Python Code

Main Idea

- Divide and Conquer
- Divide the unsorted list into n sublists, each containing one element (a list of one element is considered sorted).
- Repeatedly merge sublists to produce new sorted sublists until there is only one sublist remaining. This will be the sorted list.

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def merge(left, right):  
    # merges two lists and returns result  
  
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
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    left = ... # the left half of the list  
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    left = merge_sort(left)  
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Start sort of list

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Sort left half of list

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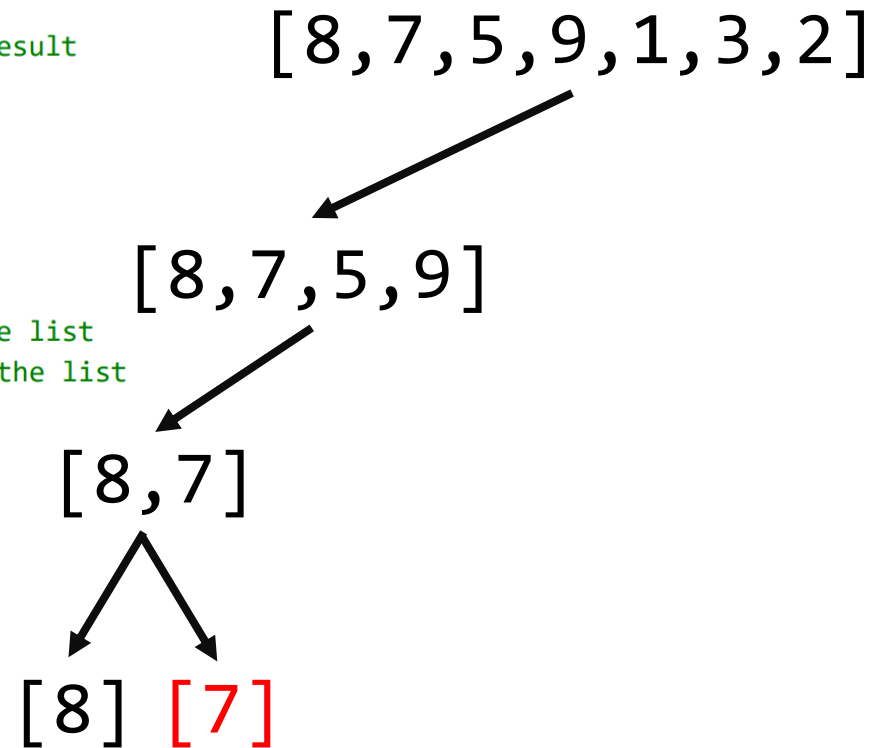
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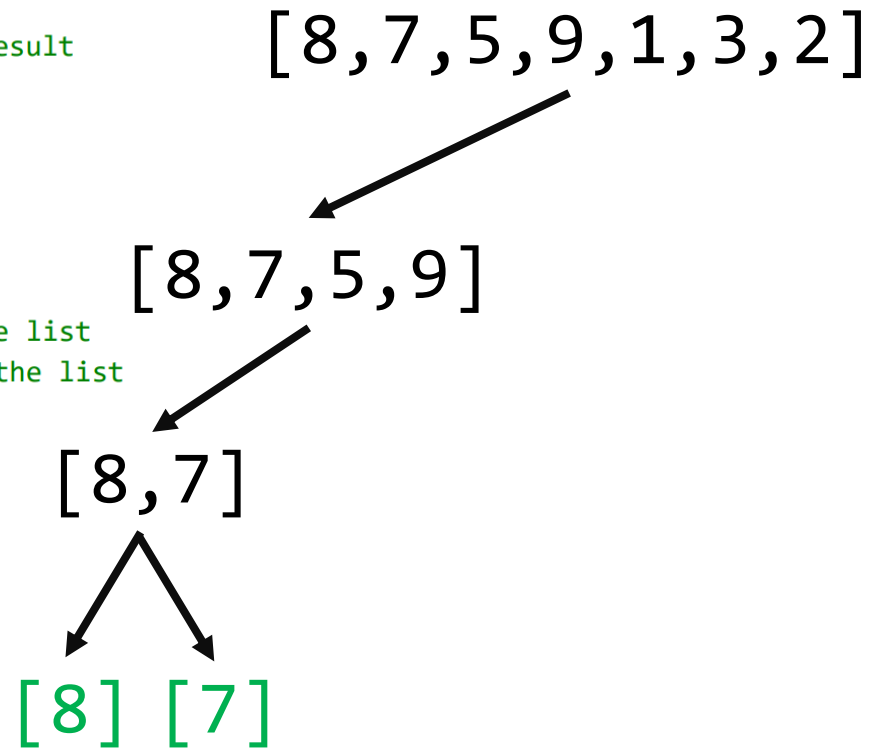
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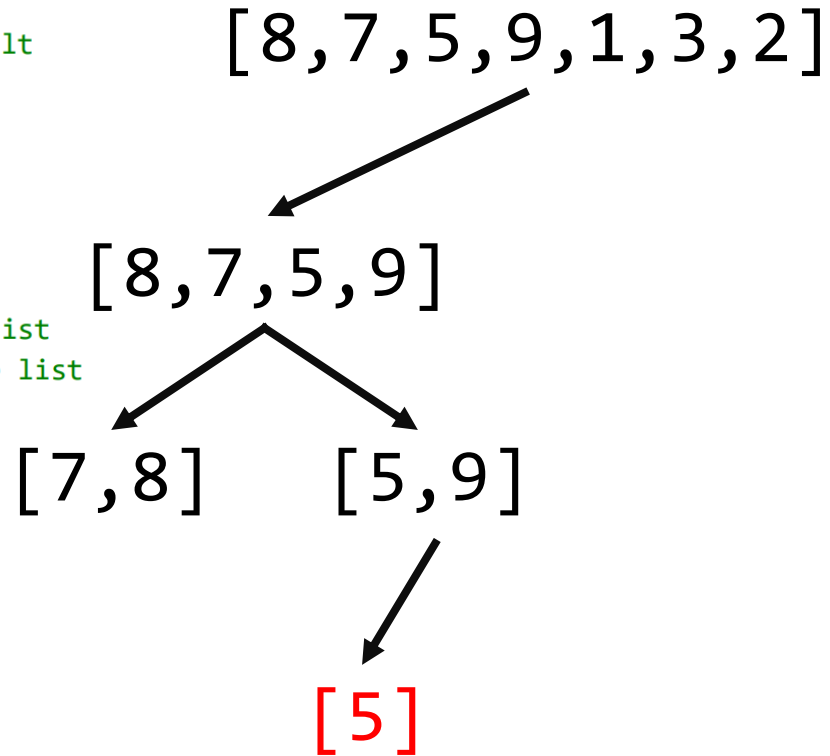
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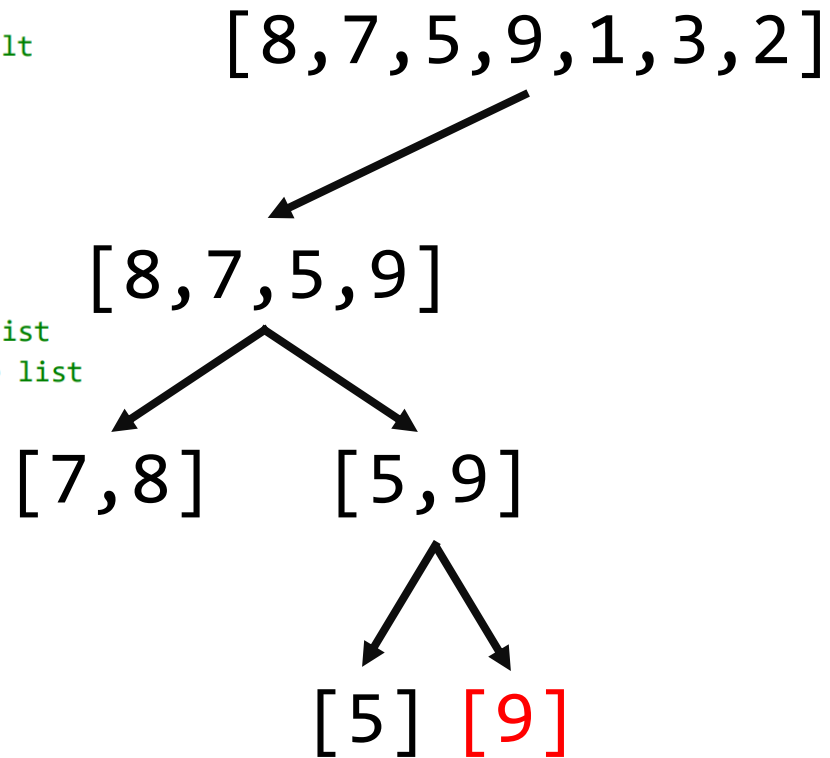
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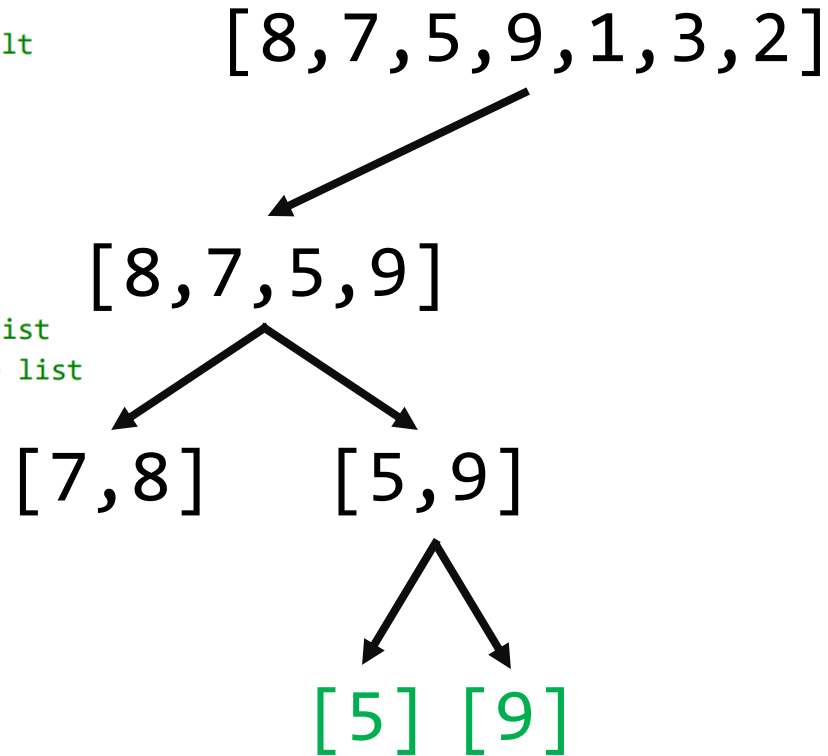
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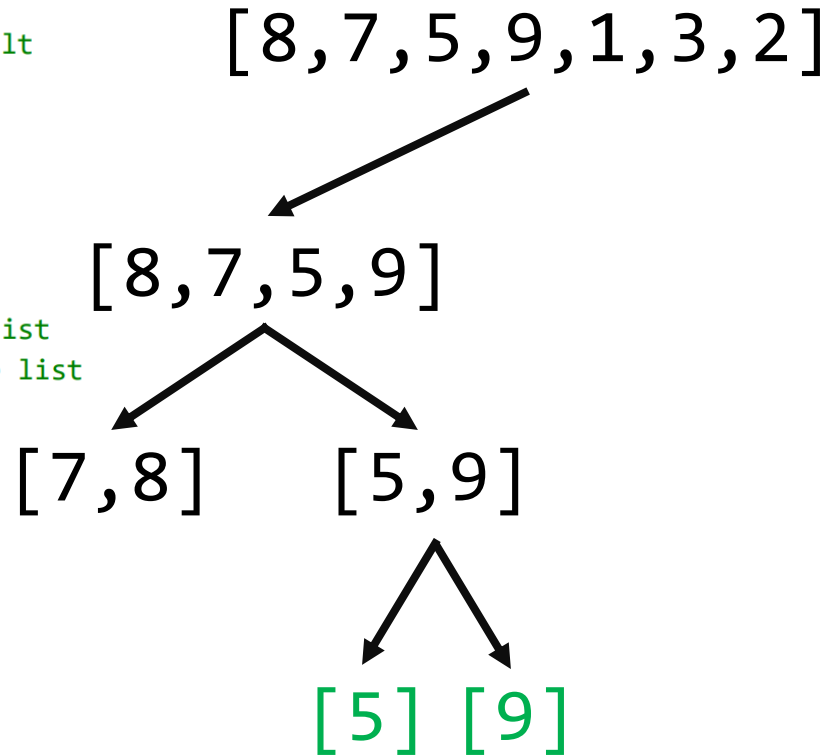


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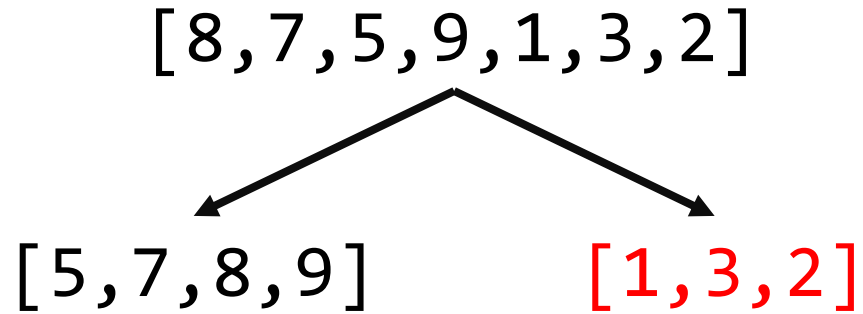
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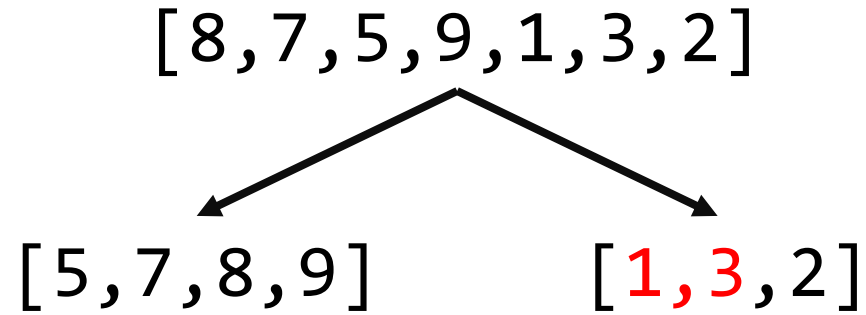
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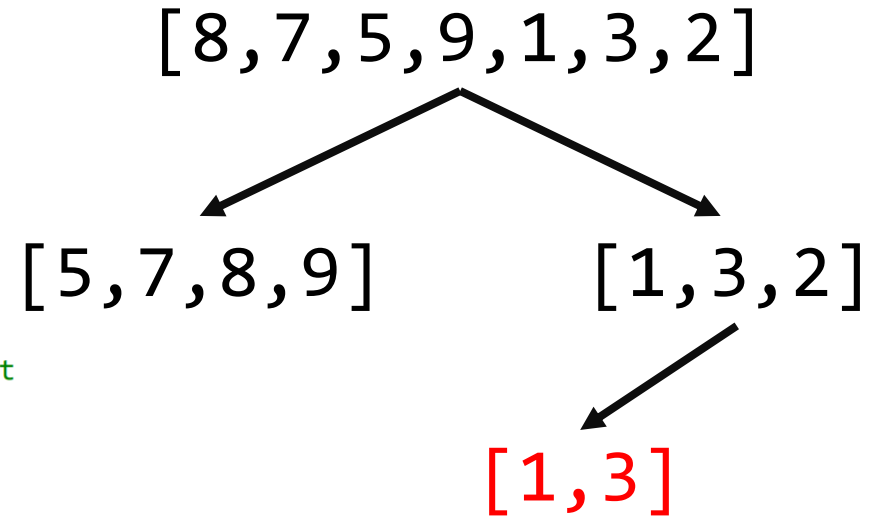
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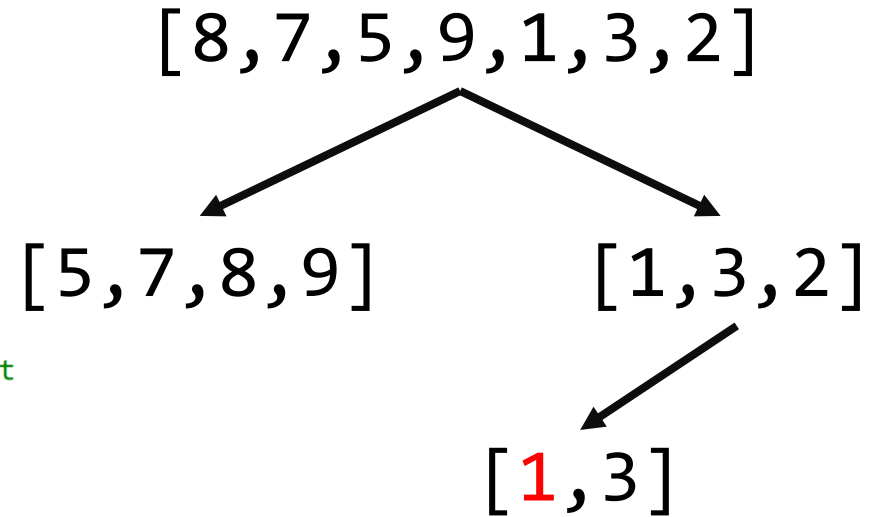
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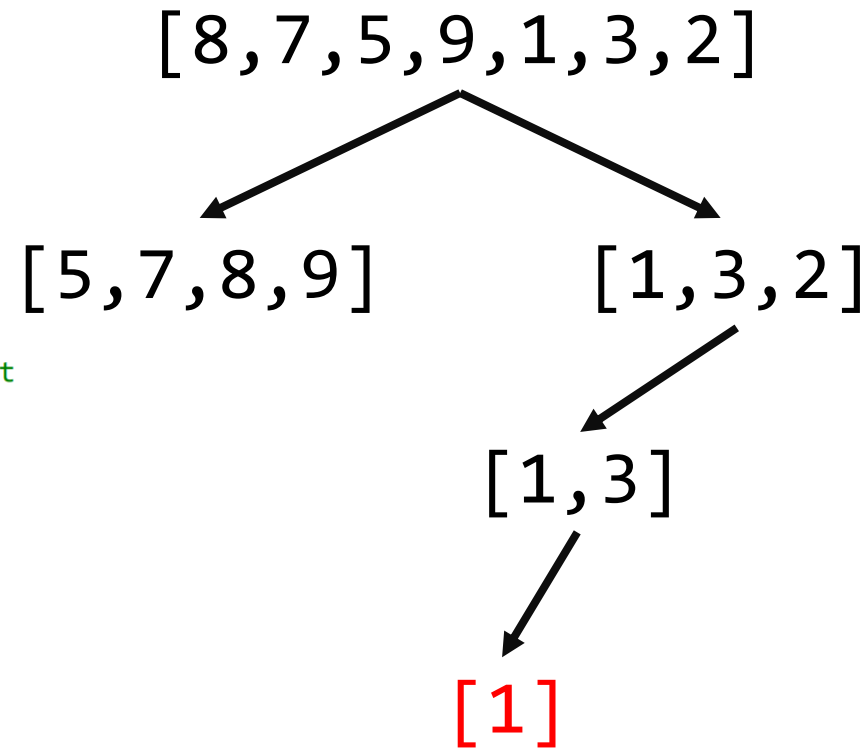
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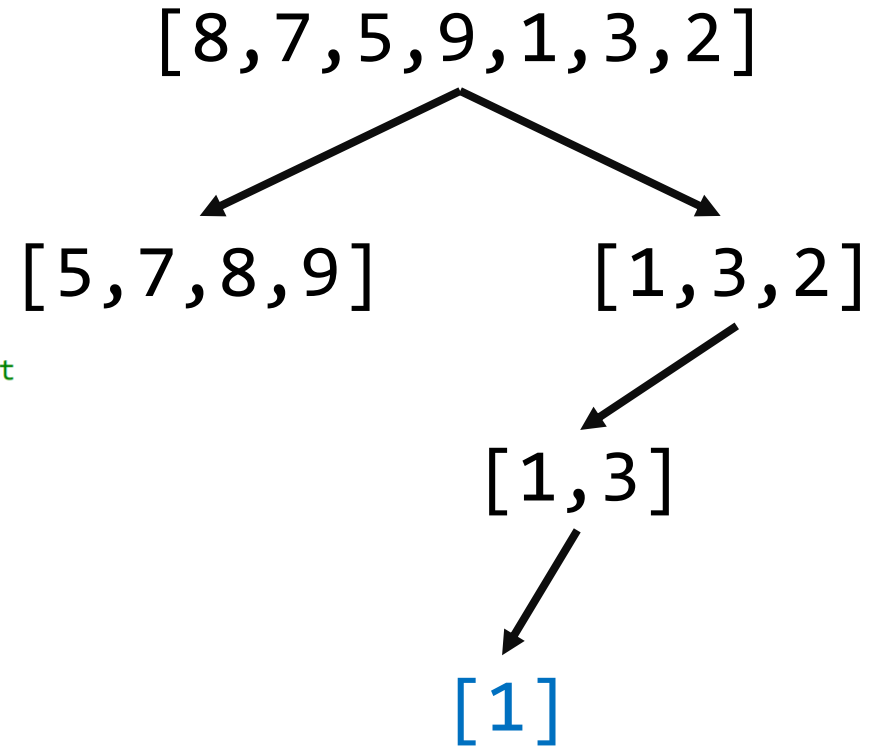
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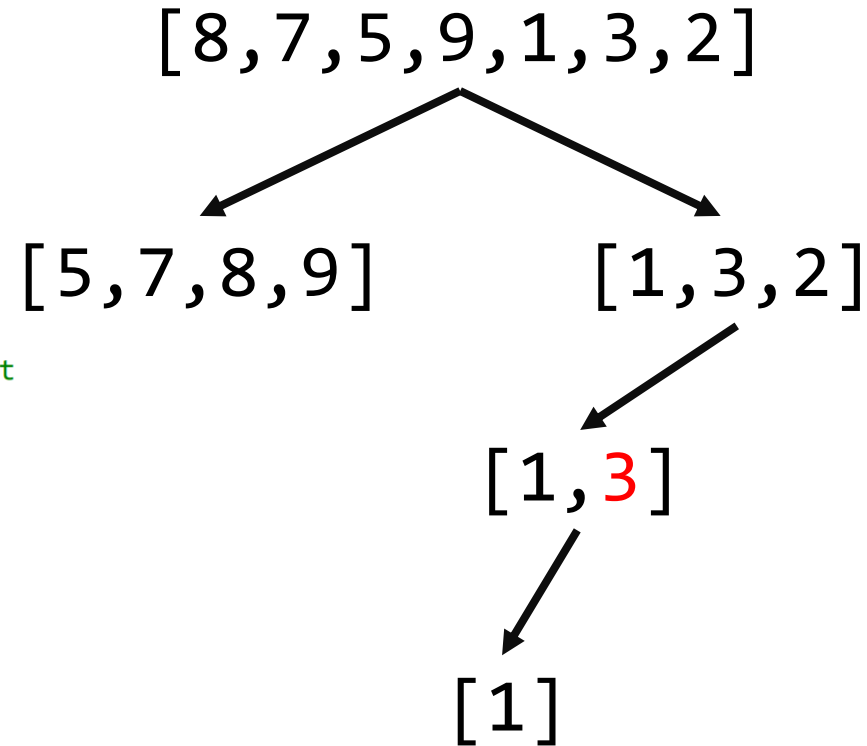
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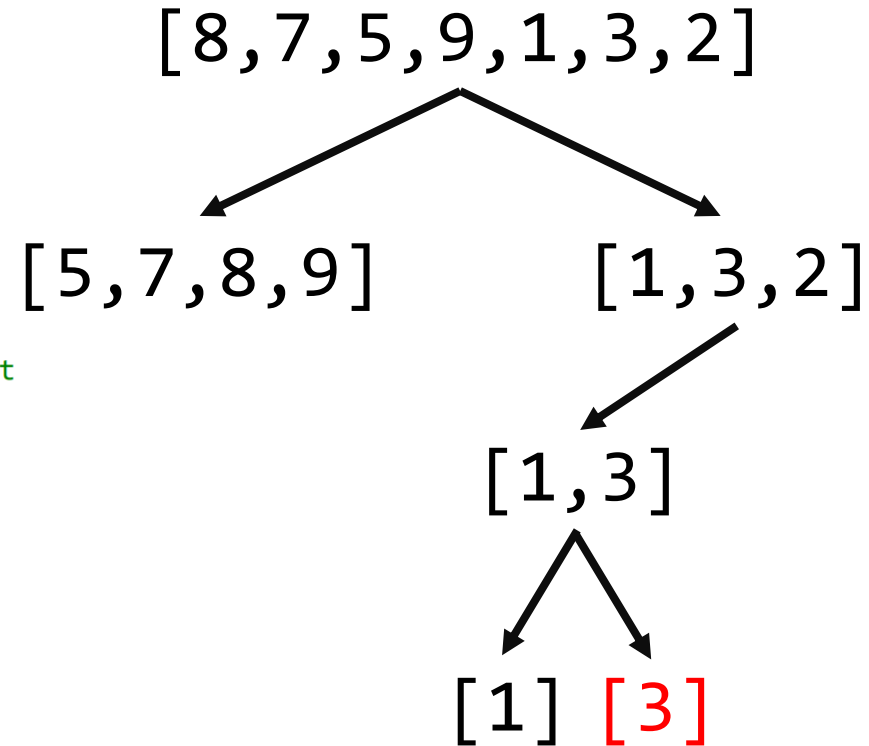
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        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



This is not valid python code, it is pseudocode...

```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):
```

```
    # base case
```

```
    if len(items) <= 1:
```

```
        return items
```

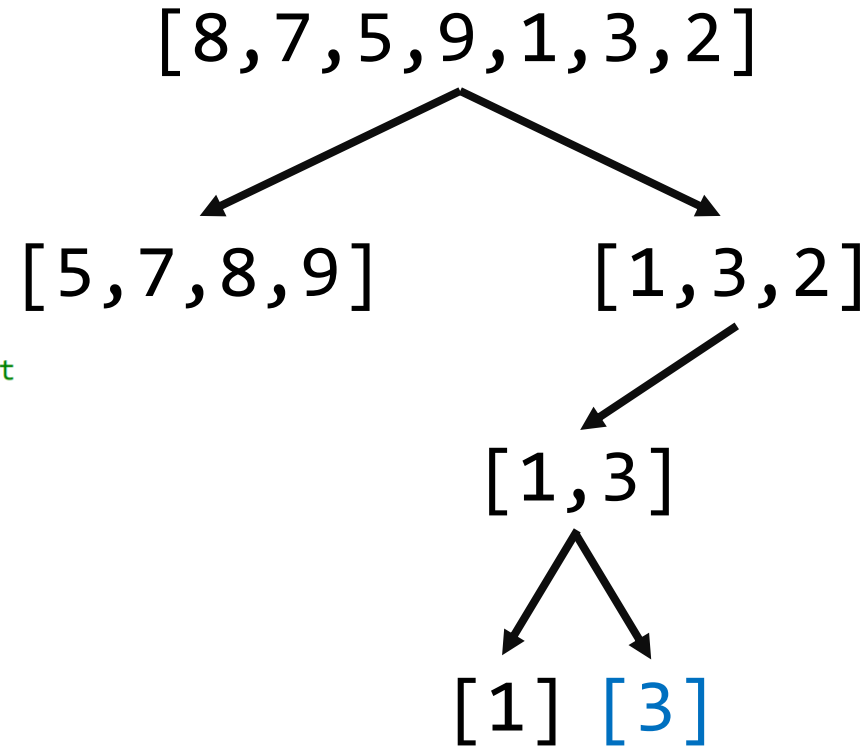
```
    left = ... # the left half of the list
```

```
    right = ... # the right half of the list
```

```
    left = merge_sort(left)
```

```
    right = merge_sort(right)
```

```
    return merge(left, right)
```



Base Case! Length of list is 1

This is not valid python code, it is pseudocode...

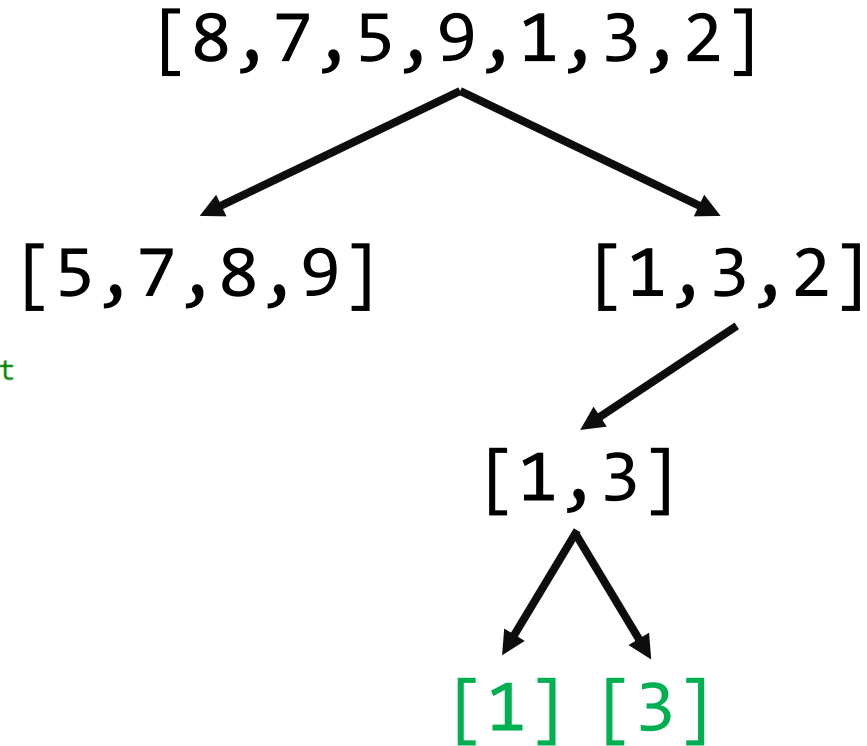
```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

→ `return merge(left, right)`

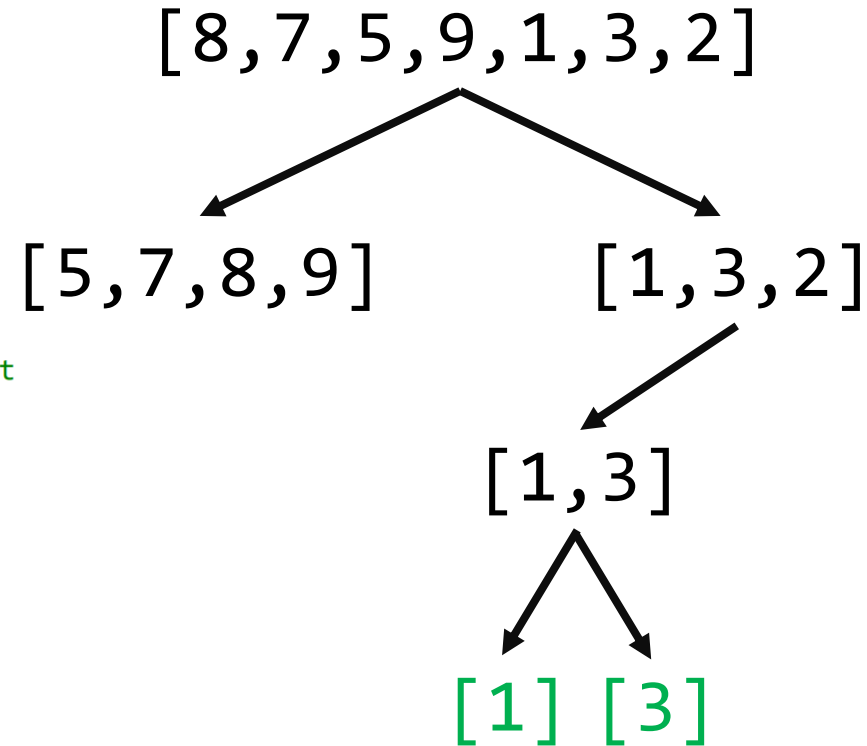


We now merge the two lists

This is not valid python code, it is pseudocode...

→ `def merge(left, right):`
merges two lists and returns result

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



We now merge the two lists

This is not valid python code, it is pseudocode...

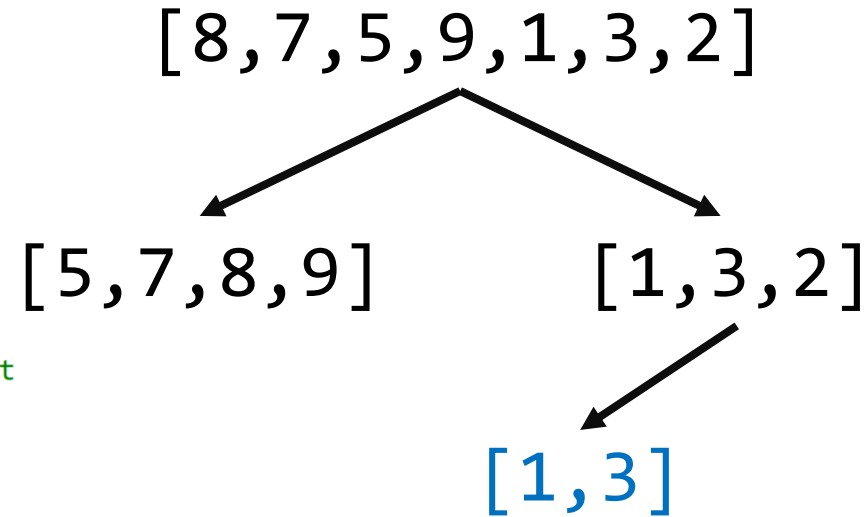
```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

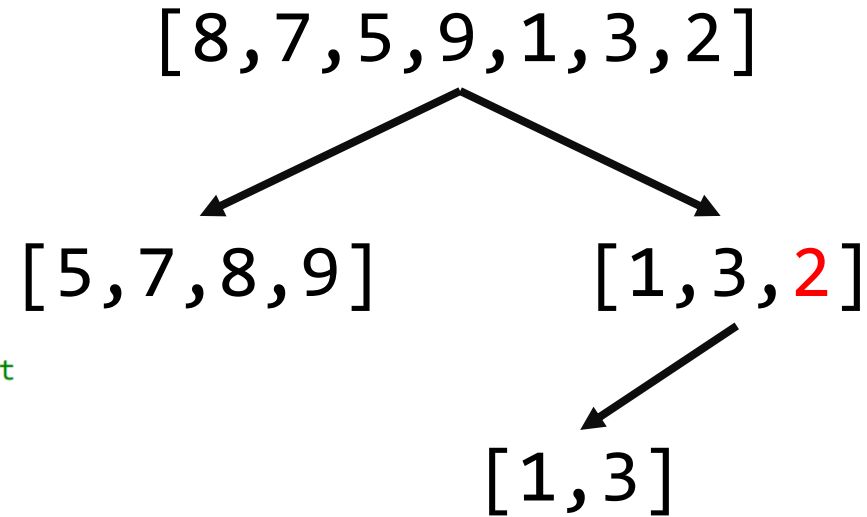
→ `return merge(left, right)`



We now merge the two lists

This is not valid python code, it is pseudocode...

```
def merge(left, right):  
    # merges two lists and returns result  
  
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



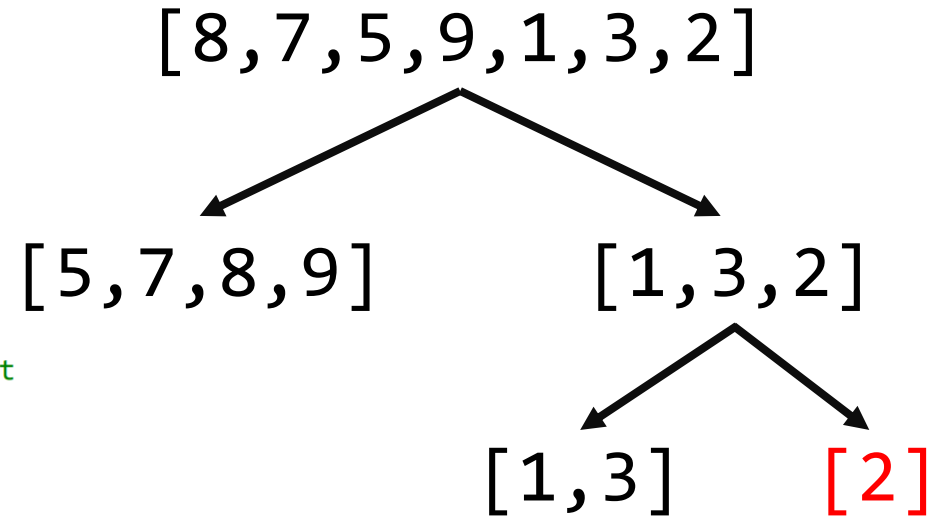
Sort right half of list

This is not valid python code, it is pseudocode...

```
def merge(left, right):  
    # merges two lists and returns result
```

→

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



This is not valid python code, it is pseudocode...

```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):
```

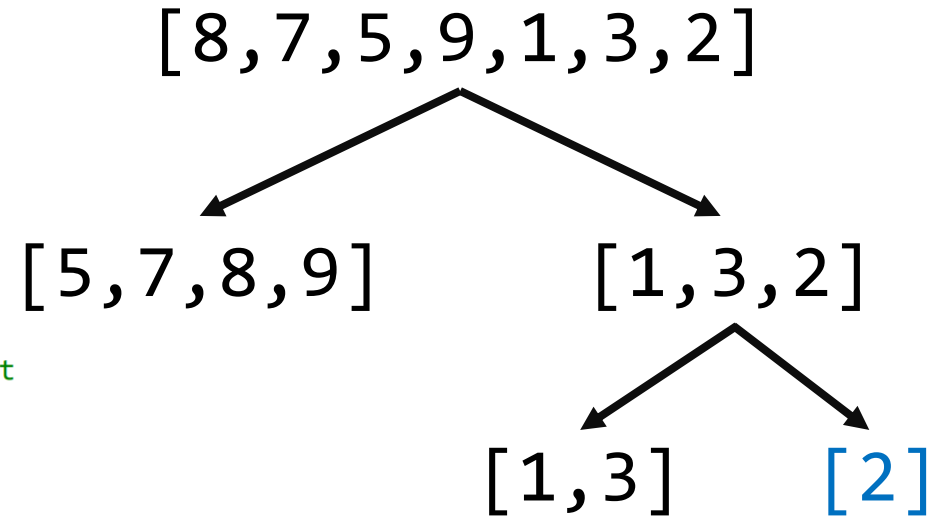
→

```
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

```
    return merge(left, right)
```



Base Case! Length of list is 1

This is not valid python code, it is pseudocode...

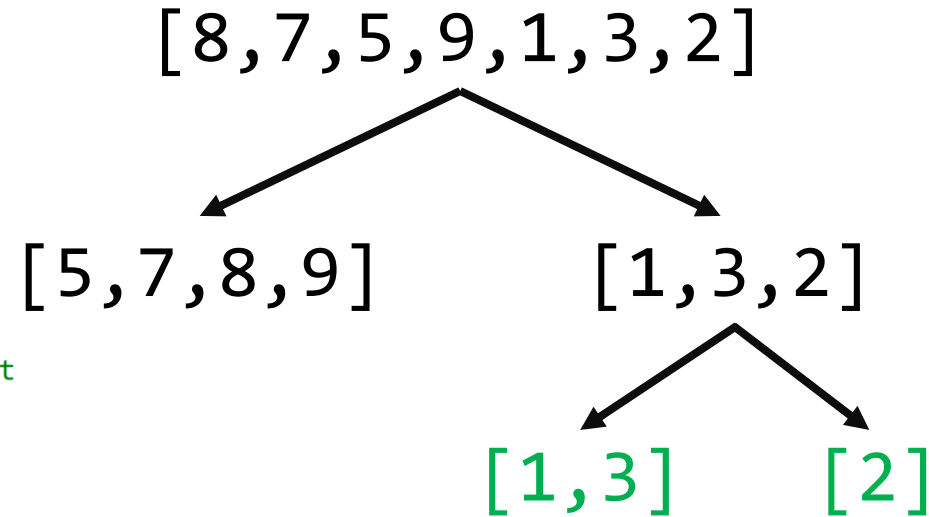
```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

→ `return merge(left, right)`

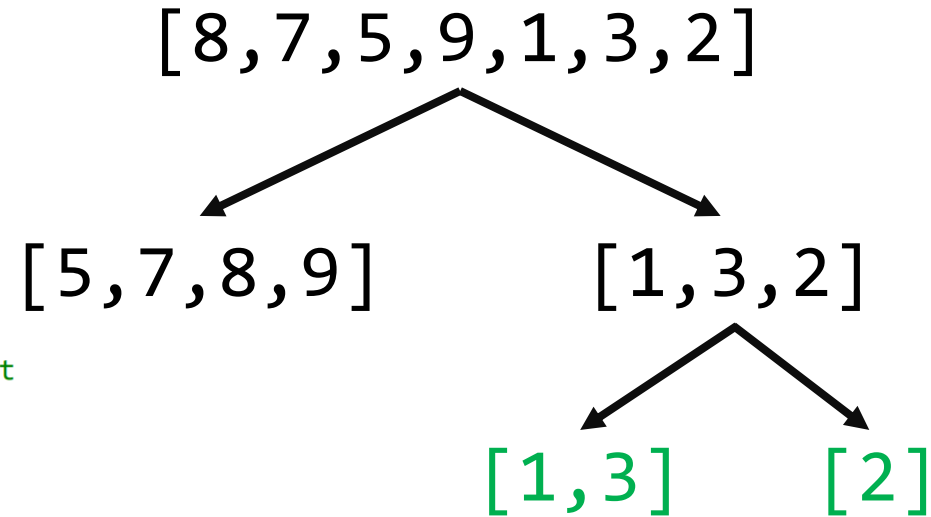


We now merge the two lists

This is not valid python code, it is pseudocode...

→ `def merge(left, right):`
merges two lists and returns result

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



We now merge the two lists

This is not valid python code, it is pseudocode...

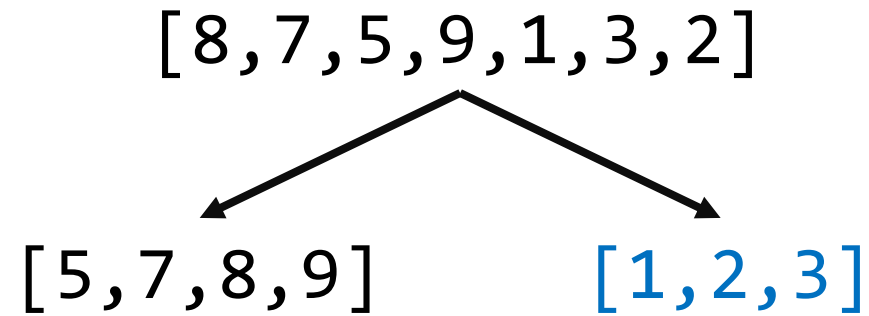
```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

→ `return merge(left, right)`



We now merge the two lists

This is not valid python code, it is pseudocode...

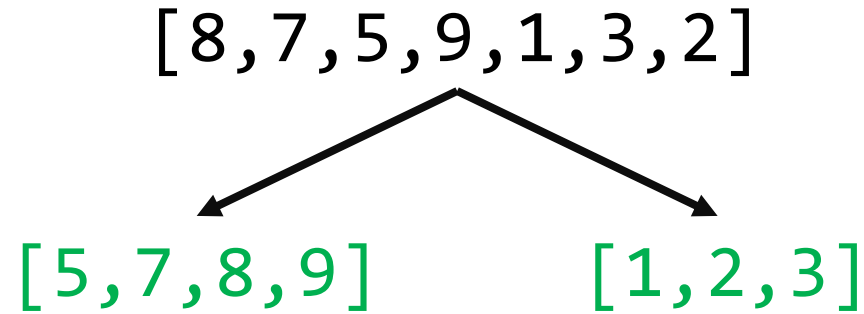
```
def merge(left, right):  
    # merges two lists and returns result
```

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items
```

```
    left = ... # the left half of the list  
    right = ... # the right half of the list
```

```
    left = merge_sort(left)  
    right = merge_sort(right)
```

→ `return merge(left, right)`

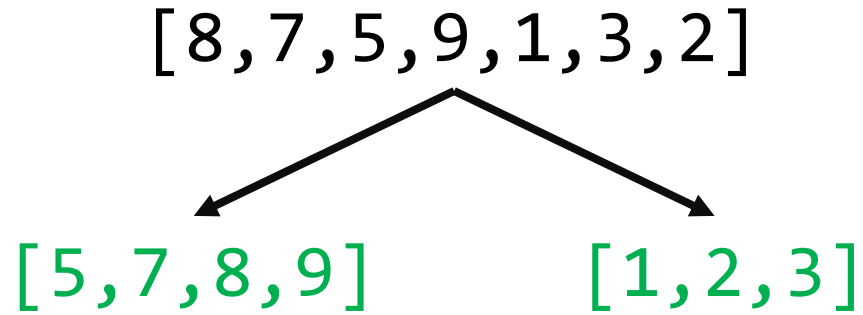


We now merge the two lists

This is not valid python code, it is pseudocode...

→ `def merge(left, right):`
merges two lists and returns result

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
  
    return merge(left, right)
```



We now merge the two lists

This is not valid python code, it is pseudocode...

```
def merge(left, right):  
    # merges two lists and returns result
```

[1,2,3,5,7,8,9]

```
def merge_sort(items):  
    # base case  
    if len(items) <= 1:  
        return items  
  
    left = ... # the left half of the list  
    right = ... # the right half of the list  
  
    left = merge_sort(left)  
    right = merge_sort(right)  
    → return merge(left, right)
```

We now merge the two lists

```
[1,2,3,5,7,8,9]
```

The list is now sorted!

Visualising the whole process

